

LIFE SCIENCE FOR MIDDLE SCHOOL (CK- 12)

Life Science for Middle School

CK-12 Foundation

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CHAPTER OVERVIEW

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1.1: Scientific Ways of Thinking



What was that noise?

If you let your emotions rule your thinking, you might think a thumping noise was a ghost. If you think like a scientist, however, you ask questions and make observations. You'd observe the shutters are loose and blowing in the wind.

Scientific Ways of Thinking

Modern science is a way of understanding the physical world, based on observable evidence, reasoning, and repeated testing. That means scientists explain the world based on their own observations. If they develop new ideas about the way the world works, they set up a way to test these new ideas.

Thinking Like a Scientist

A scientist is always trying to find the truth and discover new truths. How can you think like a scientist? Thinking like a scientist is based on asking and answering questions. Though you may not know it, you do this all day long. Scientists ask questions, and then make detailed observations to try to ask more specific questions and develop a **hypothesis**. They may design and perform an **experiment** to try to answer their question and test their **hypothesis**. From the results of their **experiment**, scientists draw **conclusions**. A conclusion describes what the evidence tells the scientist.

- **Scientists ask questions:** The key to being a great scientist is to ask questions. Imagine you are a scientist in the African Congo. While in the field, you observe one group of healthy chimpanzees on the north side of the jungle. On the other side of the jungle, you find a group of chimpanzees that are mysteriously dying. What questions might you ask? A good scientist might ask the following two questions:
 1. "What differs between the two environments where the chimpanzees live?"
 2. "Are there differences in behavior between the two groups of chimpanzees?"
- **Scientists make detailed observations:** To **observe** means to watch and study attentively. A person untrained in the sciences may only observe, "The chimps on one side of the jungle are dying, while chimps on the other side of the jungle are healthy." A scientist, however, will make more detailed observations. Can you think of ways to make this observation more detailed? What about the number of chimps? Are they male or female? Young or old? What do they eat? A good scientist may observe, "While all seven adult females and three adult males on the north side of the jungle are healthy and show normal behavior, four female and five male chimps under the age of five on the south side have died." Detailed observations can ultimately help scientists design their experiments and answer their questions. From these observations, a scientist will develop a **hypothesis** to explain the observations. A hypothesis is the scientist's proposed explanation for his or her observations. The scientist's hypothesis may be that "Young chimps on the south side die due to a lack of nutrients in their diet."

 An adult and infant chimpanzee

Figure 1.1.1: An adult and infant chimpanzee (*Pan troglodytes*).

- **Scientists find answers using tests:** When scientists want to answer a question, they search for evidence using experiments. An **experiment** is a test to see if their explanation is right or wrong. **Evidence** is made up of the observations a scientist makes during an experiment. To study the cause of death in the chimpanzees, scientists may give the chimps nutrients in the form of nuts, berries, and vitamins to see if they are dying from a lack of food. This test is the experiment. If fewer chimps die, then the experiment shows that the chimps may have died from not having enough food. This is the evidence.
- **Scientists question the answers:** Good scientists are skeptical. Scientists never use only one piece of evidence to form a conclusion. For example, the chimpanzees in the experiment may have died from a lack of food, but can you think of another explanation for their death? They may have died from a virus, or from another less obvious cause. More experiments need to be completed before scientists can be sure. Science is about finding the truth, no matter what. So good scientists constantly question their own conclusions. They also find other scientists to confirm or disagree with their evidence.

Summary

- Modern science is a way of understanding the physical world, based on observable evidence, reasoning, and repeated testing.
- To think like a scientist, you must ask questions, make detailed observations, develop a hypothesis, find answers using tests, and question your answers.

Explore More

Use the resource below to answer the questions that follow.

- **Understanding Science** at undsci.berkeley.edu/article/scienceflowchart

1. Once an experiment has been conducted and the results analyzed, what 4 possible responses are there when interpreting the data?
2. How does "peer review" fit into the scientific process? Why is it so important?
3. In the flowchart, what 5 processes are involved in "Exploration and Discovery"? Do you think any one of these processes is more important than the others? Explain your reasoning.

Review

1. What is modern science?
2. How do you think scientifically?
3. What does it mean "to observe"?
4. What is a hypothesis?

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1.2: Fields in the Life Sciences



What kind of scientist studies dolphins?

Dolphins are living organisms, so studying them is part of the life sciences. The life sciences, however, are broken down into many fields. Scientists that study dolphins and other life in the ocean are called marine biologists.

Fields in the Life Sciences

The **life sciences** are the study of living organisms. They deal with every aspect of living organisms, from the biology of [cells](#), to the biology of individual organisms, to how these organisms interact with other organisms and their environment.

The life sciences are so complex that most scientists focus on just one or two subspecialties. If you want to study insects, what would you be called? An entomologist. If you want to study the tiny things that give us the flu, then you need to enter the field of **virology**, the study of viruses. If you want to study the nervous system, which life science field is right for you (Table [below](#), Table [below](#), and Table [below](#))?

Field	Focus
Botany	Plants
Zoology	Animals
Marine biology	Organisms living in oceans
Freshwater biology	Organisms living in and around freshwater lakes, streams, rivers, ponds, etc.
Microbiology	Microorganisms
Bacteriology	Bacteria
Virology	Viruses
Entomology	Insects
Taxonomy	The classification of organisms

Field	Focus
Cell biology	Cells and their structures/functions
Anatomy	Structures of animals
Morphology	Form and structure of living organisms
Physiology	Physical and chemical functions of tissues and organs
Immunology	Mechanisms inside organisms that protect them from disease and infection
Neuroscience	The nervous system
Developmental biology and embryology	Growth and development of plants and animals
Genetics	Genetic makeup of living organisms and heredity
Biochemistry	Chemistry of living organisms
Molecular biology	Nucleic acids and proteins
Epidemiology	How diseases arise and spread
Evolution	The changing of species over time

Field	Focus
Ecology	How various organisms interact with their environments
Biogeography	Distribution of living organisms
Population biology	The biodiversity, evolution, and environmental biology of populations of organisms

During the study of the life sciences, you will study **cell biology**, **genetics**, **molecular biology**, **botany**, **microbiology**, **zoology**, **evolution**, **ecology**, and **physiology**. Cell biology is the study of cellular structure and function (Figure below). Genetics is the study of **heredity**, which is the passing of traits (and genes) from one generation to the next. Molecular biology is the study of molecules, such as DNA and proteins. Ecologists study **ecosystems**, which are made of both living and nonliving parts of the environment. A botanist may work in a botanical garden, where plant life can be studied. What will you study with the other subspecialties?

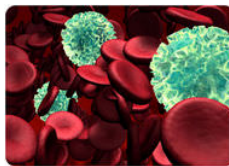


Figure 1.2.1: This illustration shows a virus among red blood cells. Which fields study red blood cells and viruses? (Keep in mind that viruses are actually much smaller than cells.)



Figure 1.2.2: Other life science subspecialties include biogeography, which is the study of where organisms live and at what abundance.

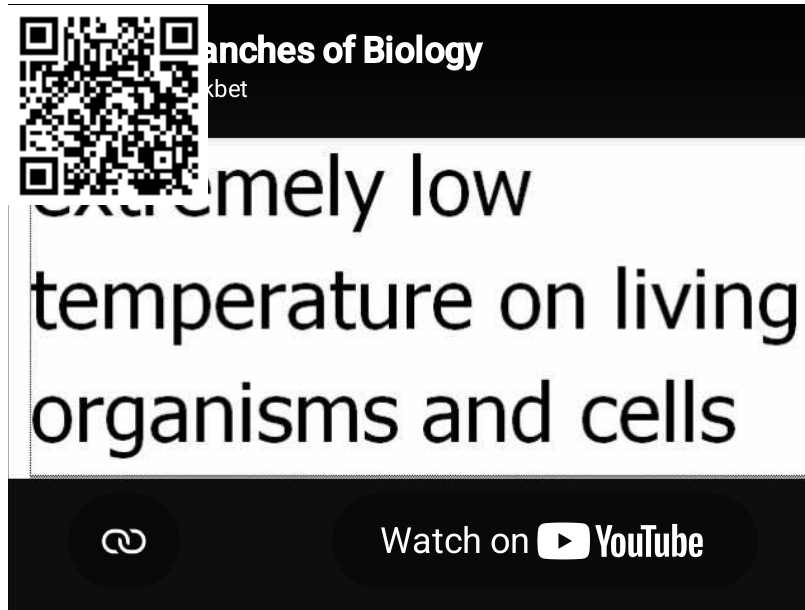
Summary

- There are several subspecialties within the life sciences that focus on one type of organism, such as virology and bacteriology.
- There are several fields of the life sciences that examine interactions between organisms and their environments, such as ecology.

Explore More

Use the resource below to answer the questions that follow.

- **Branches of Biology** at http://www.youtube.com/watch?v=OrlOOJ0Tm_E (3:28)



1. What is the study of reptiles and amphibians?
2. What is the study of prehistoric life by means of fossils?
3. What is the study of mollusks?
4. What is the study of [cells](#)?
5. What is the study of fungi?
6. What is ecology?

Review

1. What is name of the field of the life sciences that studies insects?
2. What is name of the field of the life sciences that studies the nervous system?
3. What are cell biology, genetics, and molecular biology?

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